



**Verified Carbon
Standard**

26MWP SOLAR PHOTOVOLTAIC POWER PLANT IN BAVET CITY, CAMBODIA

Document Prepared by Climate Bridge (Shanghai) Ltd.

Project Title	26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia
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1 PROJECT DETAILS

1.1 Summary Description of the Project

26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia (hereafter referred to as “the project”) is a solar power project with total installed capacity of 26MWp, located in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia. The project is invested by Green Sustainable Ventures (Cambodia) Co., Ltd. (hereafter referred to as “the project owner”).

The purpose of the project is to utilize the solar energy at the project site to generate and supply electricity to Electricite Du Cambodge Grid (EDC grid) which is dominated by coal-fired power stations. Prior to the implementation of the project, the electricity delivered to the grid by the project activity would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the EDC grid. This is the scenario that existed prior to the implementation of the project activity and is also the baseline scenario.

The project is a renewable energy project and is expected to supply an annual average of 37,085 MWh of zero-emission electricity to EDC grid during the whole project lifetime of 25 years, which replace the power generation of those fossil fuel-fired power plants delivered to the EDC grid under the baseline scenario.

The project started construction on 13-Apr-2021 and commissioned on 27-Jan-2022. The expected total emission reductions are 159,532 tCO₂e during the first crediting period from 27-Jan-2022 to 26-Jan-2029, with average annual emission reductions of 22,790 tCO₂e.

In this first monitoring period from 27-Jan-2022 to 31-Dec-2022, the emission reductions achieved by the project are 20,393 tCO₂e.

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Joint validation and verification ^{1st}	<u>(27-Jan-2022–31-Dec-2022)</u>	<u>VCS</u>	<u>CTI Certification Co., Ltd</u>	1
<u>Total</u>	<u>(27-Jan-2022–31-Dec-2022)</u>	<u>VCS</u>	<u>CTI Certification Co., Ltd</u>	1

1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy (renewable/non-renewable).

The project is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

- 1) The seven Kyoto Protocol greenhouse gases: The emissions reduction of the project comes from the sources: CO₂ emissions from the production of the equivalent amount of electricity replaced by the project that would otherwise have been purchased from the fossil fuel fired power plants of EDC grid. Thus, the project applicable to this scope.
- 2) Ozone-depleting substances: NA
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: NA.
- 4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The methodology ACM0002 (Version 21.0) of the project utilized is a methodology approved under CDM Program that is a VCS approved GHG program.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: NA.

The project utilizes the solar energy to generate and supply electricity to EDC grid which is dominated by coal-fired power plants. Baseline emissions of the project include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The project assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants, which is in line with the methodology ACM0002 (Version 21.0). Therefore, the baseline emission cannot reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

Meanwhile, Cambodia is Least Developed Country as designated by the United Nations¹, the project does not belong to the projects excluded in Table 1 of VCS Standard version 4.4.

As per the section 3.1 of VCS standard version 4.4, each clause is demonstrated as below.

Paragraph 3.1.1: The project is developing as VCS project in accordance with VCS standard version 4.4 and all procedures for registration and issuance are following Registration and Issuance Process version 4.3. The relevant templates and forms on the VCS website have been applied. The project also has applied the principles taken from ISO 14064-2 to ensure that GHG-related information is a true and fair account. Thus, the project meets all applicable rules and requirements set out under VCS Program.

¹ <https://www.un.org/en/conferences/least-developed-countries>

Paragraph 3.1.2 and 3.1.3: The latest methodology ACM0002 (Version 21.0) of the project utilized is a methodology approved under CDM Program that is a VCS approved GHG program. The application of the relevant tools referred to by this methodology is also listed in Table 3-2 of this report. The baseline scenario is assessed based on the latest methodology in the section 3.4 of this report. Thus, Paragraph 3.1.2 and 3.1.3 requirements are met by the project.

Paragraph 3.1.4: As described in the section 1.14 of this report, the project is in compliance with the local laws and not lead to the violation. Paragraph 3.1.4 requirement is met by the project.

Paragraph 3.1.5: The project does not involve model application.

Paragraph 3.1.6: The baseline scenario and additionality have been demonstrated based on the latest methodology ACM0002 (Version 21.0) in the section 3.4 and 3.5 of this report. Paragraph 3.1.6 requirement is met by the project.

Paragraph 3.1.7: There is no conflict with the rules and requirements of the VCS Program.

Paragraph 3.1.8: The application of the methodology ACM0002 (Version 21.0) and the relevant tools has been described in the section 3.2 of this report. Paragraph 3.1.8 requirement is met by the project.

Paragraph 3.1.9: The latest VCS standard version 4.4 is applied by the project.

Thus, the project is eligible under the scope of VCS program.

1.4 Project Design

The project has been designed to be a single installation of an activity, not a grouped project.

Eligibility Criteria

Not applicable as the project is not grouped project.

1.5 Project Proponent

Organization name	Green Sustainable Ventures (Cambodia) Co., Ltd.
Contact person	Charles Tan
Title	Director
Address	No.6 Street No. 588. Sangkat Boeung Kok 11, Khan Toul Kork, Phnom Penh, Cambodia.

Telephone	+60-125228603
Email	charlestan@pestech-international.com

1.6 Other Entities Involved in the Project

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86-2162462036
Email	gao.zhiwen@climatebridge.com ; projects@climatebridge.com

1.7 Ownership

The project owner of the project is Green Sustainable Ventures (Cambodia) Co., Ltd.. The shareholders of the project owner are Salah Essa and Astoria Solar Farm SDN.BHD which is wholly owned by Pestech. Astoria Solar Farm SDN.BHD owned 94% of the equity of the project owner. Salah Essa, a Cambodian individual owned 6% of the equity of project owner.

Based on the Land Lease Agreement, Salah Essa who owns the project land has leased the project land to the project owner for 20 years to operate this solar project, indicating that the project owner has the land use right to the project land.

The approval of Environmental Impact Assessment (EIA), Project Approval, and the business license of the project owner is evidence for legislative right. The EPC contract is the evidence for the ownership of the plant, equipment and power generating.

1.8 Project Start Date

27-Jan-2022 (operation start date is the date on which the project began generating electricity and the project activity began generating GHG emission reductions).

1.9 Project Crediting Period

The project adopts seven years, twice renewable crediting period for a total of 21 years, and the first crediting period is from 27-Jan-2022 to 26-Jan-2029.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	☒
Large project	☐

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
27-Jan-2022~31-Dec-2022	21,487
01-Jan-2023~31-Dec-2023	23,027
01-Jan-2024~31-Dec-2024	22,912
01-Jan-2025~31-Dec-2025	22,797
01-Jan-2026~31-Dec-2026	22,683
01-Jan-2027~31-Dec-2027	22,570
01-Jan-2028~31-Dec-2028	22,457
01-Jan-2029~26-Jan-2029	1,599
Total estimated ERs	159,532
Total number of crediting years	7
Average annual ERs	22,790

1.11 Description of the Project Activity

The project started construction on 13-Apr-2021 and commissioned on 27-Jan-2022. The total installed capacity is 26MWp which consists of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The conversion ratio of inverters from direct current (DC) to alternating

current (AC) is 1.3, i.e., $DC/AC=1.3$. Thus, the total installed capacity is $20MW_{ac}^2$ on the side of alternating current. The project lifetime of solar PV cell modules is 25 years. The main technical parameters of the key equipment of the project are illustrated in table below.

Table 1-1. The key technical parameters of solar cell modules and inverters³

Item		Unit	Technical parameters
Solar cell module	Type	-	Q. PEAK DU0 XL-G9.3 455W
	Number	-	57,639
	Rated Power	Wp	455
	Rated Voltage	V	44.34
	Rated Current	A	10.15
	Open Circuit voltage	V	53.18
	Short Circuit current	A	10.65
	Efficiency	%	≥ 20
	Lifetime	year	25
	Manufacturer	-	Hanwha Q CELLS (Qidong) Co., Ltd.
Inverter	Type	-	SG250HX
	Number	-	120
	Rated Output power	kW	225
	Rated Output Voltage	V	800
	Rated Output frequency	Hz	50/60
	Manufacturer	-	Sungrow Power Supply Co.,Ltd.

The electricity generation supplied by the project is sent to the 115/22kV Chrak Mtes Substation by the 22kV transmission line, and then delivered to the EDC grid through 115kV transmission line. The process flow chart of the project please refers to Figure 1-1. The accuracy of all meters is not lower than 0.5S.

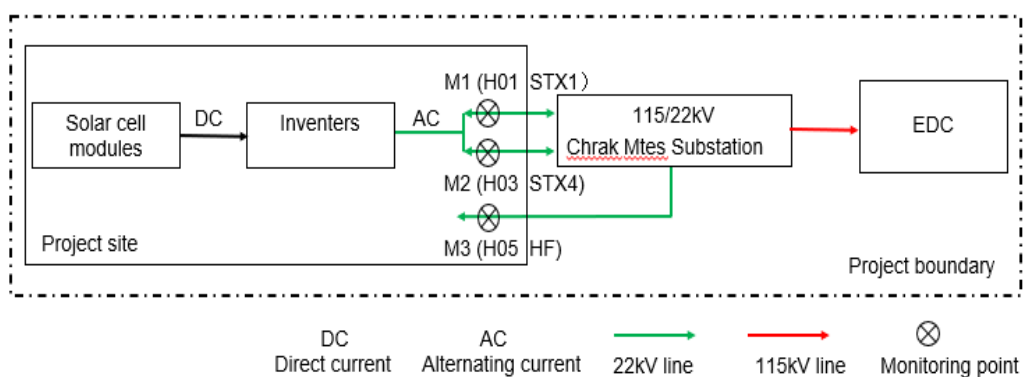


Figure 1-1. The process flow chart of the Project

² $20MW_{ac}=26MW_p/1.3$

³ Sourced from the specifications of the solar cell modules and inverters.

1.12 Project Location

The project is located in Monourom village, Bavet City, Svay Rieng Province, Cambodia. The geographical coordinates for the project site are east longitude 106°02'49.1" and north latitude 11°05'20.2". The geographic location of the project plant is shown in Figure 1-2.

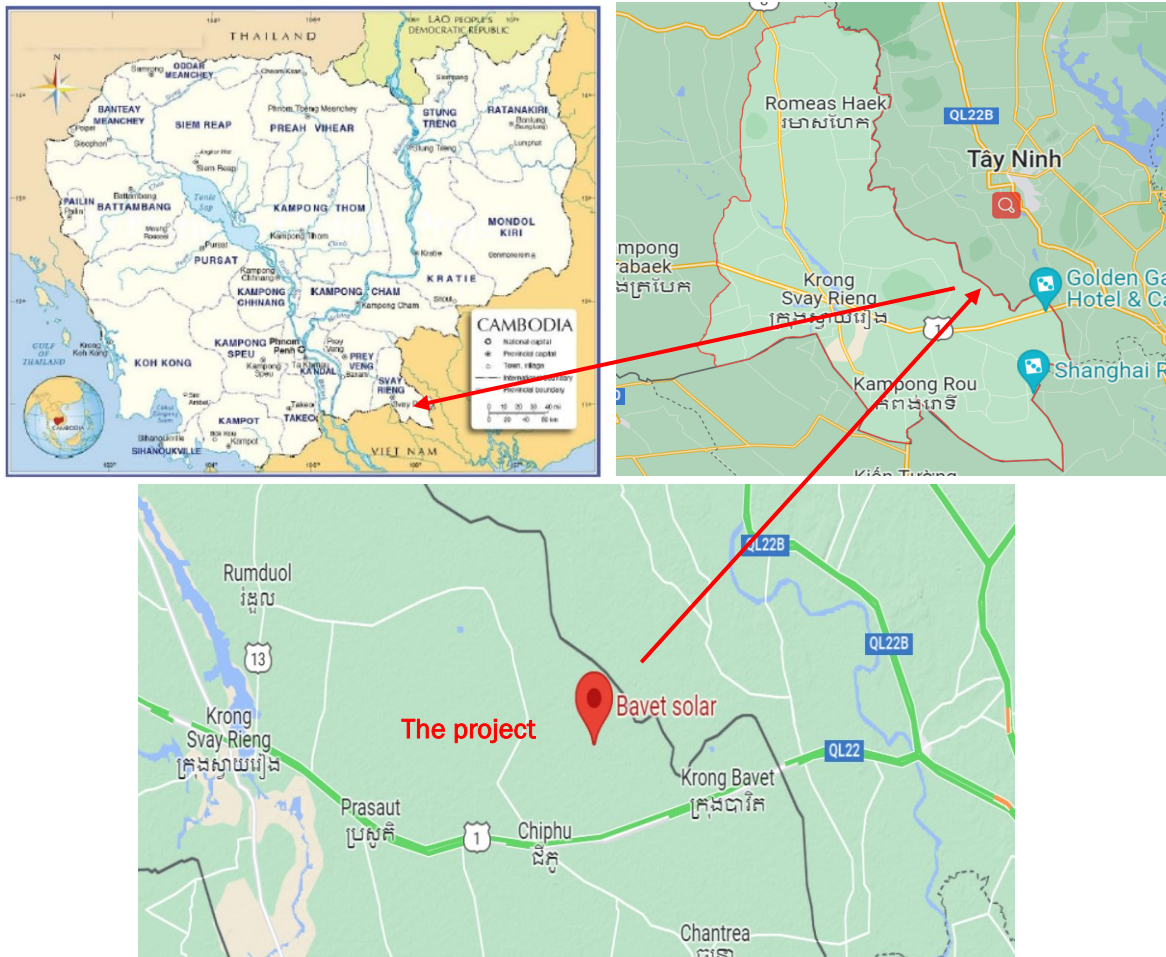


Figure 1-2 The geographic location of the project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is the electricity delivered to the EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The baseline scenario is the same as the conditions existing prior to the project initiation.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Cambodia relevant laws and regulations. Mainly include:

1. The Law on Environmental Protection and Natural Resource Management.

The Law on Environmental Protection and Natural Resource Management issued in 1996 and Ministry of Environment (MOE) was established to implement this provision. According to this provision, the project owner commissioned ERD Consultants Co., Ltd. to implement EIA for this project and obtained the EIA Approval from MOE. Thus, the project is in compliance with The Law on Environmental Protection and Natural Resource Management.

2. Electricity Law of the Kingdom of Cambodia.

The purpose of this law is to govern and to prepare a framework for, the electric power supply and services throughout the Kingdom of Cambodia. This law covers all activities related to the supply, the provision of services and uses of electricity and other associated activities of power sector. The electricity in this law shall mean the electrical energy or the capacity for the production of electrical energy. The electrical power service in this law shall mean the generation, transmission and distribution of electricity and other additional or miscellaneous services related to the above services. The project obtained the project approval from Ministry of Mines and Energy (MME) and generation license from Electricity Authority of Cambodia (EAC) which demonstrated that local government permits the construction of the project and the electric power supply and services. Moreover, the project owner also has signed Power Purchase Agreement (PPA) with EDC grid, indicated that the electric power supply and services are determined. Consequently, the project is in compliance with Electricity Law of the Kingdom of Cambodia.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered nor seeking registration under any other GHG programs.

The project is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

Not applicable. The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Cambodia does not have a national emissions trading scheme⁴. Moreover, the project owner promised that the emission reductions generated by the project will not be double counting.

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project hasn't sought or received another form of environmental credits.

1.16.3 Supply Chain (Scope 3) Emissions

The project is grid-connected renewable solar power project and the project proponent built and operated the power plant to generate the renewable electricity which is supplied to the EDC grid. The project does not involve retailers. Therefore, the project proponent promised that the emission reduction generated by the project will only seek VCUs and will not be double counting.

As per the VCS Clarification 1⁵, projects are not required to complete the sections in the affected VCS project templates that relate to Scope 3 emissions double claiming until the effective date of the revised requirements of 01-Jan-2024. Therefore, this section is not applicable.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project is a solar power project with total installed capacity of 26MWp, located in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia.

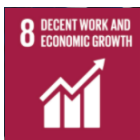


The project will utilize solar resource to generate electricity delivered to the EDC grid which would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the EDC grid. The project will achieve a GHG emission reduction of 22,790 tCO₂e/yr during the first crediting period. Thus, the project will achieve SDG 13 Climate Action⁶.

⁴ <https://disruptiveasia.asiasociety.org/the-growth-of-carbon-markets-in-asia>

⁵ <https://verra.org/wp-content/uploads/2023/06/VCS-Clarification-on-Scope-3-Emissions-Double-Claiming-Requirements-31-May-2023.pdf>

⁶ <https://sdgs.un.org/goals/goal13>



The construction and operation of the project will provide 7 employment positions for local villagers, which will have a positive effect on the local economy. During first crediting period, direct and indirect employment opportunities will be generated. Thus, the project will achieve SDG 8 Decent Work and Economic Growth⁷.



38,773 MWh/yr of electricity from renewable sources will be supplied to EDC grid in the first crediting period. Thus, the project will achieve SDG 7 Affordable and Clean Energy⁸.

1.17.2 Sustainable Development Contributions Activity Monitoring

The project utilized solar resource for power generation, in this monitoring period, the project has generated 34,694.918 MWh of electricity exported to EDC grid and achieved 20,393 tCO₂e emission reduction, which also offered 7 employment positions for local villagers. Therefore, the project achieved SDG 7, SDG 8 and SDG 13 in this monitoring period.

⁷ <https://sdgs.un.org/goals/goal8>

⁸ <https://sdgs.un.org/goals/goal7>

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	34,694.918 MWh of electricity from renewable sources have been supplied to the EDC grid during the first monitoring period, replacing fossil fuel powered electricity.	From 27-Jan-2022 onwards, 34,694.918 MWh of electricity from renewable sources has been supplied to the EDC grid.
2)	8.3	Provide job opportunities by implementation of the project	Implemented activities to increase	Provided 7 employment positions for local residents. No further changes in the first monitoring period.	Provided 7 employment positions for local residents from 27-Jan-2022 onwards.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The project achieved a GHG emission reduction of 20,393 tCO ₂ e during the first monitoring period.	From 27-Jan-2022 onwards, the project has achieved GHG emission reduction of 20,393 tCO ₂ e.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable as the project does not involve any additional information that may have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

The EIA Report for the project has been approved by Ministry of Environment on 30-Nov-2021. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development, and no net harm has been detected. Please refer to section 2.3 for details. Especially for site preparation and civil works, the project owner has planted grass under solar photovoltaic panels to prevent water loss and soil erosion during operation period.



Figure 2-1 The picture of grass under solar photovoltaic panels

Meanwhile, the implementation of the project will improve local socio-economic development through creating career opportunities and paying taxes.

Though the land is sugar plantation prior to construction, no employment on plantations is reported among major income sources, indicating that plantation work is seasonal or an additional livelihood. Additionally, overwhelming majority of the project area households live mainly on farming. A small number of households also run small businesses. A considerable number of people earn their living as workers in garment industry and as construction workers. Therefore, the project replaces sugar plantation with little impact on the income of local residents. Besides, Salah Essa purchased the project land from previous landowners and then leased it to the project owner for 20 years to operate this solar project. All previous landowners had received total land sold fees and there is no economic dispute. Thus, there are no negative impacts on local socio-economy.

In conclusion, the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

Local Stakeholder Consultation during the project preparation stage:

As the entire plot of land was sugar plantation (image appended below) to which the previous owners have agreed to sell to Salah Essa, there is no resettlement concern. Salah Essa leased the land to the project owner for 20 years to operate this solar project. Furthermore, Cambodia

Government has strict laws in place on land acquisition and involuntary settlement in place to provide assurance. Prior to construction, the project has acquired the construction permit issued by Ministry of Land Management on 13-Apr-2021.



Figure 2-2 The landscape of sugar plantation

Prior to the construction, the project owner has prepared a stakeholder analysis and communication strategy as a guide for meaningful information disclosure, consultation and active participation of all project stakeholders. The stakeholders of the project mainly involved local village chiefs and villagers.

The project owner entrusted ERD Consultants Co., Ltd. to visit local village chiefs at the project site in November 2020 and collect some data by questionnaire as per the stakeholder analysis and communication strategy. The local village chiefs understood and supported the project and had no comments.

During the construction process, local villagers not only got job opportunities, but also got a better understanding of the project. Most local stakeholders believe that the project can improve their living standards and promote local economic development without too much adverse environmental impact. At the same time, based on Health Safety Environment monthly report during the construction period, there were no social conflicts or safety accidents at the project site, further indicating that the villagers are very supportive of the project and the construction of civil works.

In conclusion, the stakeholders are supportive to the project and the construction of civil works, believing that the project will provide more employment opportunities, help local economic development, and increase local power supply. The implementation of the project is regarded as beneficial by most of the local stakeholders.

Local Stakeholder Consultation during the project implementation stage:

Communications with local stakeholders are being carried out at periodic intervals which is every quarter. Key implementation schedules or changes of the project will be communicated to the

neighbourhood committee and the local residents, the comments and suggestions from residents will be collected by the project owner and considered during the implementation phase. A grievance signage shown in Figure 2-4 has been placed out the project gate for anyone to provide feedback. The relevant grievance redress mechanism is listed as following.

The project staff should conduct friendly contact and discussions with relevant stakeholders to explain the appeal; secondly, for any grievances that are not resolved by amicable negotiations, the project manager should propose a solution and mediation which is performed by relevant authority based on all collected information; Thirdly, for any grievances that are not resolved through mediation, if necessary, it will be addressed by relevant legal method such as arbitration and courts.

In this first monitoring period, the project owner conducted physical outreach to the neighbouring communities and documented community issues in Monthly Operation & Maintenance Report.

In February 2022, the project owner visited 6 of villagers, 1 of village chief and 6 of the project workers face to face to collect their comments, all of them expressed satisfaction with the project. Besides, everyone can provide feedback to the project owner via phone or email written on the grievance signage.



Figure 2-3 The interview with local villagers

In March 2022, the maintenance team of the project went to trim the trees to avoid tripping over the power transmission line, but the nearby Pagoda Monks directly prevented the maintenance team from trimming. Since the first item of the grievance redress mechanism could not solve the problem, the project manager invited the village chief to discuss with the Monk head and

explained that the tree branches that were too high would trip the transmission line and cause problems such as electric shock. After mediation, the Monk head understood the importance of pruning the branches and agreed to do so. There is no negative comment received for the project in other months during this monitoring period.

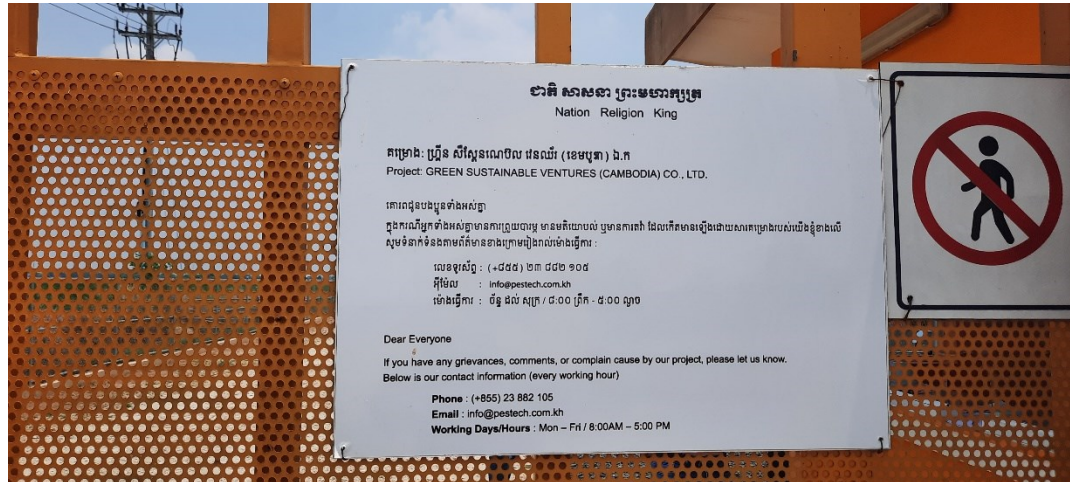


Figure 2-4 A grievance signage

2.3 Environmental Impact

The final EIA report of the project was approved by Ministry of Environment on 30-Nov-2021, indicating that EIA is in compliance with The Law on Environmental Protection and Natural Resource Management. The environmental impacts of the project are summarized as follows.

Topography (Land and vegetation)

As the project site has already been cleared, the site preparation involves a limited amount of vegetation clearance. Site preparation and civil works will change the land use. However, prior to construction, the project has acquired the construction permit issued by Ministry of Land Management on 13-Apr-2021. The procedures of land acquisition are in compliance with the Land Management in Cambodia.

Vegetation clearances are strictly restricted to the construction sites temporary areas and access roads. No chemicals (herbicides) are used. Loose soil from foundations is restored through ramming. Excess soil is spread out in areas such that it does not disturb the local drainage pattern. Engineering and biological measures are implemented to prevent soil erosion such as sow soil binding grass. Landscaping/tree plantation with native species along the project site fenced area will be done as soon as earthworks are completed to stabilize the soil and reduce dust pollution.

During the operation period, periodic vegetation maintenance (e.g. trimming of shrubs, grass and tree branches) will be undertaken at the project site and within the transmission right of way in order to maintain necessary safety clearances. Manual vegetation maintenance methods will be used, and local labor will be utilized where possible creating a potential job opportunity.

Air pollution

In the construction period, opened and exposed land areas at the project site, drilling activities, loose dust at the base of the panel structures, are sprayed with water to suppress dust levels particularly the dry season. Aside from this, the construction site is temporarily enclosed to control dust level. Construction vehicles transporting materials that generate dusts are covered with tarps. During the operation period, the project does not involve exhaust gas emissions.

Water pollution

In the construction period, the EPC contractor provided workers with sanitary facilities to treat domestic wastewater or sewage and complied with National Sub-decree No.27 on Water Pollution Control.

The wastewater generated during the operation period is solar panels cleaning wastewater. The solar panels cleaning wastewater mainly includes dust, which does not require chemical treatment. Domestic wastewater can be recycled and reused for gardening. Sewage is directed to onsite system associated with sanitation facilities.

Noise

In the construction period, the use of equipment, construction vehicles and civil works may increase the noise levels such as during excavation works at the project site and induce vibration, which impact buffer area on temporary basis.

During the operation period, the main noise is generated by inverters and transformers. Operation of noise-generating equipment are enclosed, and periodic maintenance of equipment will be conducted. Noise level measurements has been conducted to ensure that noise level is complied with International Finance Corporation Environmental, Health, and Safety Guidelines⁹.

Solid waste

A covered sorting area for recyclables is established on-site in the construction period. Wastes such as packaging materials, meta and wires, electronics, wood (if any), are transported to EDC grid depots/dedicated storage yards, for resale and auction to authorized dealers. Multiple waste containers are also be provided. Waste burning in and around the site including buffer area are prohibited.

In the operation period, the solid waste generated after the decommissioning is waste equipment and waste solar panels, licensed companies can be hired to collect, transport, and dispose of unused hazardous materials and wastes.

In conclusion, the environmental impact during the project construction and operation period will be minor. The Project owner has taken appropriate measures to minimize adverse environmental impacts.

⁹ <https://www.ifc.org/content/dam/ifc/doc/2023/ifc-general-ehs-guidelines.pdf>

2.4 Public Comments

This project was open for public comment from 17-Jan-2023 to 16-Feb-2023 on the verra website. No negative comment has been received.

2.5 AFOLU-Specific Safeguards

Not applicable as the project is non-AFOLU.

3 APPLICATION OF METHODOLOGY

3.1 Project Boundary

According to ACM0002 (Version 21.0), the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the EDC grid that the project power plant is connected to. The project boundary is shown in Figure 3-1 below.

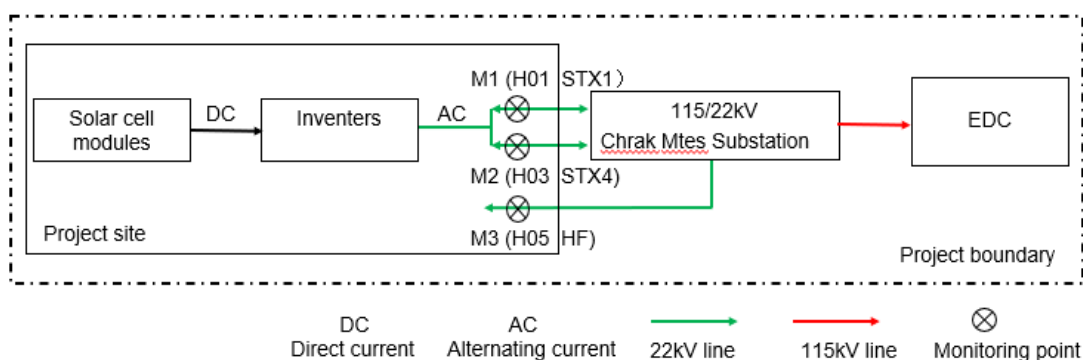


Figure 3-1 Diagram of the project boundary

The main emission sources and gases included in the project boundary is listed in table below:

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	The major source of emissions in the baseline.
	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.

Source	Gas	Included?	Justification/Explanation
displaced due to the project activity	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
Project	CO ₂	No	According to the ACM0002, the production and operation of solar photovoltaic projects do not produce significant greenhouse gas emissions. Thus, the project emissions can be ignored.
	CH ₄	No	
	N ₂ O	No	

3.2 Title and Reference of Methodology

Applied approved CDM methodology: ACM0002 Grid-connected electricity generation from renewable sources (Version 21.0)

This methodology also refers to the latest approved version of the following tools:

“Tool to calculate the emission factor for an electricity system” (version 07.0)

“Tool for the demonstration and assessment of additionality” (version 07.0.0)

“Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 3.0)

“Common practice” (version 03.1)

“Investment analysis” (version 12.0)

For more detail information about the methodology and tools, please reference to the following link: <https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQCOPiWPGWDN8ED5PG>

3.3 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology ACM0002 (Version 21.0), of which the detailed description is listed in Table 3-1 below:

Table 3-1 Applicability of ACM0002

Para No.	Applicability criteria	Justification
Scope	This methodology applies to grid-connected renewable energy generation project activities that include: (a) Construction and operation of a Greenfield power plant; or (b) Retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant.	Applicable. The project is a greenfield solar PV power plant connected to EDC grid.

	Further, the methodology applies to grid-connected renewable energy generation project activities which integrate Battery Energy Storage System (BESS) to a Greenfield power plant or to an existing solar photovoltaic or wind power plant.	The project belongs to condition (a). The project does not involve the integration of a BESS.
1	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant. (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units. (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	Applicable. The project is a greenfield solar PV power plant. The project belongs to condition (a).
2	In case the project activity involves the integration of a Battery Energy Storage System (BESS), the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	Not applicable. The project does not involve the integration of a BESS.
3	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;	Applicable. The project is a greenfield solar PV power plant. The project activity does not involve capacity additions, retrofits, rehabilitations or replacement, and the integration of a BESS. The project belongs to condition (a).

	(c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
4	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply. (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²;	This condition is not relevant, as the project activity is not the installation of a hydro power plant.

	(ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity. (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project	
5	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	This condition is not relevant, as the project activity is not the installation of a hydro power plant.
6	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units;	The project activity does not involve any of the given criteria hence methodology is applicable for the project activity.
7	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	This condition is not relevant, as the project activity does not involve capacity additions, retrofits, replacement or rehabilitations.

In addition, the project meets the applicability conditions of the applied tools applied in the PD as follows:

Table 3-2 Applicability of applied tools

Tool	Applicability criteria	Justification
Tool for the demonstration and assessment of additionality (Version 07.0.0)	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Applicable The methodology selected for the proposed project requires the use of this tool.
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Applicable The methodology applied in this proposed project requires the use of this tool.
Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer. (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the	Applicable The electricity consumed by the project is from the EDC grid, which is Scenario A.

	electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid. (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable The electricity generated in the project scenario is supplied to the EDC grid, which is Scenario I.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	Applicable No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.
Tool to calculate the emission factor for an electricity system (Version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable This project replaces grid power supply and uses this tool to calculate the values of OM, BM and CM of this project.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available	Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO ₂

	to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by this tool. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia. To be conservative and accurate, the grid emission factor $EF_{grid,CM,y}$ was 0.5878 tCO₂/MWh which is from the latest IFI Dataset of Default Grid Factors v.3.2. Please refer to section 3.6 for details.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	This condition is not relevant, there is no part of the power system of this project located in Annex I countries.
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	This condition is not relevant, this project is a solar power project.
Investment analysis (Version 12.0)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis	Applicable The project applies the methodological tool “Tool for the demonstration and assessment of additionality”.

	for the demonstration of additionality and/or the identification of the baseline scenario	
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology ACM0002 (Version 21.0) applied in this project requires the use of this tool to demonstrate the investment analysis of this project.
Common practice (Version 03.1)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Applicable This project applies the methodological tool “Tool for the demonstration and assessment of additionality”.
	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology ACM0002 (Version 21.0) applied in this project requires the use of this tool to demonstrate the common practice of this project.

3.4 Baseline Scenario

According to the methodology ACM0002 (Version 21.0), as the project is the installation of a Greenfield solar power plant, the baseline scenario of the project is electricity delivered to EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

3.5 Additionality

As per the relevant applicable laws and regulations at validation, i.e. Electricity Law of The Kingdom of Cambodia¹⁰, Long-term strategy for Carbon Neutrality¹¹, Cambodia Basic Energy Plan¹² the project activity is in compliance with all laws and regulations in Cambodia and the project activity is not mandated by any law, statute, or other and regulations.

¹⁰ [https://portal.mrcmekong.org/assets/v1/documents/Cambodian-Law/-Electricity-Law-\(2001\).pdf](https://portal.mrcmekong.org/assets/v1/documents/Cambodian-Law/-Electricity-Law-(2001).pdf)

¹¹ <https://unfccc.int/documents/417790>

¹² https://www.eria.org/uploads/media/CAMBODIA_BEP_Fullreport_1.pdf

According to the methodology ACM0002 (Version 21.0), the additionality of the project activity shall be demonstrated and assessed using the latest version of the “TOOL01: Tool for the demonstration and assessment of additionality”. The steps are listed as following.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Define realistic and credible alternatives to the project activity(s) through the following Sub-steps.

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participant or similar project developers that provide outputs or services comparable with the project activity are identified below:

Alternative (a) The proposed project activity undertaken without being registered as a VCS project activity.

Alternative (b) Continuation of the current situation (no project activity or other alternatives undertaken).

Sub-step 1b: Consistency with mandatory laws and regulations

According to Cambodia Basic Energy Plan¹³ prepared by The General Department of Energy, Ministry of Mines and Energy of Cambodia (MME), the MME will open the power generation market to local and foreign entities such as the Asian Development Bank and private companies to install solar/PV in Cambodia for supplying electricity to local people. There is no legal prohibition regarding the implementation of the project currently or in the future, therefore alternative (a) is in compliance with mandatory laws and regulations.

Alternative (b) Continuation of the current situation is that electricity delivered to EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. As per the Electricity Law of The Kingdom of Cambodia¹⁴, there are no laws or regulations in Cambodia that prohibit the power grid from providing the same annual electricity generation. Therefore alternative (b) is in compliance with mandatory laws and regulations.

Thus, both alternatives (a) and (b) are realistic and credible alternatives in Cambodia to the project which are consistent with mandatory laws and regulations.

Step 2: Investment analysis

Sub-step 2a: Determine appropriate analysis method

¹³ https://www.eria.org/uploads/media/CAMBODIA_BEP_Fullreport_1.pdf

¹⁴ [https://portal.mrcmekong.org/assets/v1/documents/Cambodian-Law/-Electricity-Law-\(2001\).pdf](https://portal.mrcmekong.org/assets/v1/documents/Cambodian-Law/-Electricity-Law-(2001).pdf)

According to the Tool for the demonstration and assessment of additionality” (version 07.0.0), if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

For the project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) is used.

Sub-step 2b: Option III. Apply benchmark analysis

According to Tool for the demonstration and assessment of additionality” (version 07.0.0), “Local commercial lending rates are appropriate benchmarks for a project IRR”. The local lending rate offered by National Bank of Cambodia (NBC) is chosen by the project entity for investment analysis. Since the PPA was signed by both project owner and the EDC grid on 27-August-2019 which is considered as the investment decision date, Cambodian average commercial bank prime lending rate in 2018 (10.61%¹⁵) is chosen as the benchmark for the project investment analysis.

Based on above benchmark, calculation and comparison of financial indicators are carried out in sub-step 2c.

Sub-step 2c: Calculation and comparison of financial indicators

The project owner of the project is Green Sustainable Ventures (Cambodia) Co., Ltd.. The shareholders of the project owner are Salah Essa and Astoria Solar Farm SDN.BHD which is wholly owned by Pestech. Astoria Solar Farm SDN.BHD owned 94% of the equity of the project owner. Salah Essa, a Cambodian individual owned 6% of the equity of project owner.

aNew Energy, as energy investment expert, introduced Astoria Solar Farm SDN.BHD which is wholly owned by Pestech to establish the project owner with Salah Essa together. aNew Energy¹⁶ originates and co-develops projects related to wind, solar and energy storage with their partners and their clients. aNew Energy also assists in merger and acquisition, fund raise, and market connect in Asia for energy related companies. aNew Energy specializes in structuring projects with their partners. aNew Energy strives to match investors risk-reward profiles to the relevant projects.

Financial investment analysis sheet prior to the construction was finished by Camillus Yang of aNew Energy in May 2019. Camillus has a passion for integrated renewable energy projects and matchmaking complimenting investors and technology to projects. He has most renewable energy experience and enjoys developing an options pool (structuring, financing, technology) for

¹⁵ NBC Annual Report 2018 issued on 30-Jan-2019, <https://www.nbc.org.kh/english/index.php>

¹⁶ <https://www.anew.energy/>

clients and partners alike. Therefore, the professionalism and accuracy of this financial investment analysis sheet can be guaranteed. Moreover, the completion date of this financial investment analysis sheet is very close to the investment decision date, indicating this financial investment analysis sheet is reasonable as an investment basis.

The project owner made an investment decision on 27-August-2019 to develop the project as a GHG emission reduction project such as VCS project based on this financial investment analysis sheet. This investment decision was made before the starting date of the project, indicating that VCS was seriously considered in this decision prior to the implement of the project.

Due to the impact of loan interest on income tax calculations, it is recommended that when a project IRR is calculated to demonstrate additionality a pre-tax benchmark be applied. Therefore, pre-tax project Internal Rate of Return (IRR) is calculated in investment analysis. The pre-tax project IRR is compared to the benchmark of 10.61% defined in sub-step 2b.

The main financial parameters and IRR of the project are shown in Table 3-3 below.

Table 3-3 Main parameters for financial analysis calculation

Item	Unit	Value	Source
Installed capacity	MWp	26	Financial investment analysis sheet prior to the construction of the project
Average annual operating hours	Hours	1,877	
Total static investment (excluding VAT or other taxes in Cambodia)	USD	24,718,000	
Average Annual electricity delivered to the grid	MWh/year	37,544	
Electricity Tariff (excluding VAT)	USD/MWh	76 ¹⁷	
Depreciation period	Year	20 ¹⁸	
Depreciation rate	%	5 ¹⁹	

¹⁷ As per the Power Purchase Agreement (PPA) signed by the project owner and the EDC grid, electricity tariff prior to the commercial date is 42 USD/MWh, 76 USD/MWh of electricity tariff is fixed until the end of the term of the PPA and the highest tariff that they received by supplying the electricity to the EDC grid.

¹⁸ As per Cambodian 2018 Tax Booklet, the annual depreciation charges can be calculated on the useful life by using the straight-line method. Since the useful life of this solar project is 20 years according to the PPA, the depreciation period is 20 years.

¹⁹ <https://www.pwc.com/kh/en/tax/cambodian-2018-tax-booklet1.pdf>

Annual operation and management (O&M) cost	USD/year	398,972	
Construction period	Year	1	
Operation period	Year	20	
Expected VCU Price	USD/tCO ₂ e	10	

Table 3-4 below shows the difference between the project IRR, with and without the revenue from carbon.

Table 3-4 IRR results with and without carbon credits

Item	Without carbon revenue	Benchmark rate	With carbon revenue
Project IRR (before tax)	7.88%	10.61%	9.09%

It is calculated that the pre-tax project IRR without the carbon revenues is 7.88% which is lower than the 10.61% of benchmark rate. This demonstrates that the proposed project activity is not a commercially viable option to supply power. Though the pre-tax project IRR with the carbon revenues is 9.09% which is still lower than the benchmark, the return on investment for the project with carbon revenues can be improved to a certain extent. Therefore, the pre-tax IRR with carbon revenues of 9.09% is acceptable to the project owner.

Sub-step 2d: Sensitivity analysis

The sensitivity analysis is to show whether the conclusion regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.

According to “Investment analysis” (version 12.0), only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. For the project, the total static investment constitutes more than 20% of total project costs; the product of electricity tariff and Annual electricity delivered to the grid constitute more than 20% of the total revenue of the project. And the total O&M throughout the project lifetime accounts for more than 20% of the project cost.

Therefore, considering the reasonable variations in the critical assumptions, the sensitivity analysis is made regarding the 4 key factors of total static investment, electricity tariff, annual O&M cost and annual electricity delivered to the grid, as shown in Table 3-5.

Table 3-5 Sensitivity analysis of IRR for the project.

Hypothetic change	-10%	-5%	0	+5%	+10%
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Total Static investment	9.49%	8.66%	7.88%	7.17%	6.51%
Annual electricity delivered to the grid	6.26%	7.08%	7.88%	8.67%	9.44%
Annual O&M cost	8.10%	7.99%	7.88%	7.78%	7.67%
Electricity tariff	6.26%	7.08%	7.88%	8.67%	9.44%

Table 3-6 below shows the necessary change in the four major parameters required for the Project IRR without VCU's revenue to reach the benchmark rate, ceteris paribus. All the speculated changes are clearly too large to be realistic.

Table 3-6 Critical Point Analysis

Hypothetic Project IRR	Total static investment	Annual electricity delivered to the grid	Annual O&M cost	Electricity tariff
Benchmark Rate 10.61%	-16.00%	+17.83%	-132.48%	+17.83%

Total static Investment

The Project IRR would reach the benchmark rate of 10.61% when the total static investment decreases by 16.00%. According to the investment details by the project owner, the actual investment of the project is 24,595,722 USD which is almost same as the total static investment. Therefore, it is impossible that the total static investment can be reduced by 16.00% to make the IRR to cross the benchmark.

Annual electricity delivered to the grid

The Project IRR would reach the benchmark rate of 10.61% when annual electricity delivered to the grid increased by 17.83%. The electricity delivered to the grid in the first year is calculated as 39,359²⁰ MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp. By checking the Global Solar Atlas website²¹, the annual photovoltaic electricity yield of the project is reasonable. As per solar PV module specifications provided by manufacturer Q-Cells, the annual degradation rate is 0.5%. From the second year onwards, annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV

²⁰ 39,359 MWh=26MWp*1,514 kWh/kwp, the transmission line loss can be ignored because the electricity meters are installed in the solar power plant,

²¹ <https://globalsolaratlas.info/map?c=11.119117,105.837922,10&s=11.088123,106.066388&m=site>

module, ie $39,162 \text{ MWh} = 39,359 * (1 - 0.5\%)$. Therefore, an increase of 17.83% for annual electricity delivered to the grid is basically impossible.

Annual O&M cost

Variations in the annual O&M costs have an insignificant impact on the return of the project. The IRR cannot cross benchmark when the O&M cost is zero, and it is impossible to happen at all.

Electricity tariff

The Project IRR would reach the benchmark rate of 10.61% when annual electricity tariff increased by 17.83%. According to PPA of the project and sales receipts, electricity tariff is 76 USD/MWh which is fixed until the end of the term of the PPA and the highest tariff that they received by supplying the electricity to the EDC grid. Therefore, it is not like for the project to get 17.83% increase of electricity tariff to cross the benchmark.

In conclusion, the investment analysis and sensitivity analysis show that the project undertaken without carbon credit revenue is not financially attractive and the carbon revenue will improve the financial indicators of the Project.

Step 3: Barrier analysis

The proposed project does not employ the barrier analysis. Then proceed to Step 4.

Step 4: Common practice analysis

As the project applies power generation based on renewable energy which is one of the measure(s) listed in the definitions section above, proceed to Sub-step 4a.

Sub-step 4a: The proposed project activity(ies) applies measure(s) that are listed in the definitions section above

Common practice analysis is carried out as per “Common practice” (version 03.1).

Sub-Step 4a-1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity

The install capacity of the project is 26 MWp (20 MWac), so the applicable capacity range as +/- 50% of the total design capacity is 13 ~ 39 MWp (10 ~ 30 MWac).

Sub-Step 4a-2: identify similar projects which fulfil all of the following conditions:

- (a) The projects are located in the applicable geographical area;
- (b) The projects apply the same measure as the proposed project activity;
- (c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;

- (d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;
- (e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 4a-1;
- (f) The projects started commercial operation before the project design document is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Applicable Geographical Area: The applicable geographical area is adopted to be Kingdom of Cambodia.

Measure: Project activity falls under the option (b) Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies (example: energy efficiency improvements, power generation based on renewable energy). The project uses solar to generate electricity.

Output: As per the definition of output, “electricity” is considered as output of the project activity. As per the Step 4a-1, the applicable capacity range of the project is 13 ~ 39 MWp (10 ~ 30 MWac).

Different technologies: The project activity uses solar for the generation of power and hence as per the above definition, the technology which use energy source/ fuel or feed stock other than solar (i.e. coal, wind, hydropower etc) will be considered as the different technology and hence excluded.

Commercial operation date: As the date of the project began generating GHG emission reductions is 27-Jan-2022, the starting date of the proposed project is 27-Jan-2022.

Therefore, the operational solar power plants with the capacity from 13 MWp (10 MWac) to 39 MWp (30 MWac) in Cambodia before 27-Jan-2022 are identified as similar projects.

There are no operational solar power plants with capacity from 13 MWp (10 MWac) to 39 MWp (30 MWac) in Cambodia before 27-Jan-2022 based on available published sources²².

Sub-step 4a-3: *Within the projects identified in Sub-step 4a-2, identify those that are neither registered CDM/VCS/GS project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all} .*

According to **Sub-Step 4a-2**, there are no operational solar power plants with capacity from 13 MWp (10 MWac) to 39 MWp (30 MWac) in Cambodia before 27-Jan-2022, $N_{all} = 0$.

²² https://www.edc.com.kh/Annually_page/annuallyReport

Sub-step 4-4: Within similar projects identified in Sub-step 4-3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

For the project, $N_{diff}=0$.

Sub-step 4-5: Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Since $N_{all} = 0$, $N_{diff}=0$, therefore, $N_{all} - N_{diff} = 0$, which is less than 0.

$F=1-N_{diff}/N_{all}=1$.

According to common practice (version 03.1), the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3. However, the $N_{all}-N_{diff}$ is less than 3.

Thus, the Project activity is not a “common practice”.

Outcome of Step 4:

Based on the analysis above, the baseline scenario of the project is:

Electricity delivered to EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Therefore, the proposed project is additional.

3.6 Methodology Deviations

Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia.

By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries.

According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”²³, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM)

²³ https://unfccc.int/sites/default/files/resource/IFITWG_Methodological_approach_to_common_dataset.pdf

as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is accurate and authoritative.

According to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022²⁴, the latest $EF_{grid,CM,y}$ of solar project and firm energy project (e.g. Hydro, geothermal) in Cambodia in 2021 is 0.8743 tCO₂e/MWh and 0.5878 tCO₂e/MWh, respectively. To be conservative, 0.5878 tCO₂e/MWh can be used to calculate emission reduction for the project.

The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals. The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The project started construction on 13-Apr-2021 and commissioned on 27-Jan-2022. The total installed capacity is 26MWp which consists of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The conversion ratio of inverters from direct current (DC) to alternating current (AC) is 1.3, i.e., DC/AC=1.3. Thus, the total installed capacity is 20MWac²⁵ on the side of alternating current. The project lifetime of solar PV cell modules is 25 years.

Please refer to section 1.11 for detailed technological information.

The electricity generation supplied by the project is sent to the 115/22kV Chrak Mtes Substation by the 22kV transmission line, and then delivered to the EDC grid through 115kV transmission line. The process flow chart of the project please refers to Figure 1-1. The cut-off time of electricity meters is at 24:00 last day of every month. The accuracy of all electricity meters is not lower than 0.5S.

During this first monitoring period, the quantity of electricity supplied by the project plant to the grid is 34,813.293 MWh, the quantity of electricity delivered to the project plant from the grid is 118.375 MWh, the quantity of net electricity generation supplied by the project plant to the grid is 34,694.918, and the achieved emission reduction is 20,393 tCO₂e.

The project has operated without any accidental or emergency events that might impact the applicability of the methodology, the GHG emission reductions or removals and monitoring.

²⁴ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

²⁵ 20MWac=26MWp/1.3

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The GHG emission reduction calculation of the project was based on the applied methodology ACM0002 (Version 21.0).

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

Where:

BE_y	=	Baseline emission in year y (tCO ₂ e)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM-project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (tCO ₂ e/MWh)

The project is a Greenfield solar power plant, then:

$$EG_{PJ,y} = EG_{facility,y} \quad (2)$$

Where:

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

Following ACM0002 (Version 21.0), the baseline emission factor should be calculated as combined margin emission factor of EDC grid ($EF_{grid,CM,y}$), consisting of the combination of operating margin emission factor of EDC grid ($EF_{grid,OM,y}$) and build margin emission factor of EDC grid ($EF_{grid,BM,y}$) according to the following six steps defined in the “Tool to calculate the emission factor for an electricity system” (version 7.0).

However, based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia²⁶.

Thus, according to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022²⁷, the $EF_{grid,CM,y}$ of 0.5878 tCO₂e/MWh are applied in the project and fixed in the first crediting period. Please refer to the section 3.6 for details.

The electricity delivered to the grid in the first year is calculated as 39,359²⁸ MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp. The average operation hours are 1,877h²⁹. By checking the Global Solar Atlas website³⁰, the annual photovoltaic electricity yield of the project is reasonable. As per solar PV module specifications provided by manufacturer Q-Cells, the annual degradation rate is 0.5%. From the second year onwards, annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module. The annual electricity supplied to the grid $EG_{facility,y}$ is shown in the table below. The $EF_{grid,CM,y}$ of the project is 0.5878 tCO₂e/MWh, therefore baseline emission of the project is calculated as follows:

Year	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (t CO ₂ e/MWh)	BE_y (t CO ₂ e)
	A	B	C=A*B
27-Jan-2022~31-Dec-2022	36,555	0.5878	21,487
01-Jan-2023~31-Dec-2023	39,176	0.5878	23,027
01-Jan-2024~31-Dec-2024	38,980	0.5878	22,912
01-Jan-2025~31-Dec-2025	38,785	0.5878	22,797
01-Jan-2026~31-Dec-2026	38,591	0.5878	22,683
01-Jan-2027~31-Dec-2027	38,398	0.5878	22,570
01-Jan-2028~31-Dec-2028	38,206	0.5878	22,457
01-Jan-2029~26-Jan-2029	2,721	0.5878	1,599
Total	271,411		159,532

In conclusion, the total estimated baseline emission of the project in the first crediting period is 159,532 tCO₂e.

²⁶ <https://www.moe.gov.kh/>

²⁷ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

²⁸ 39,359 MWh=26MWp*1,514 kwh/kwp, the transmission line loss can be ignored because the electricity meters are installed in the solar power plant.

²⁹ 1,877h=37,544MWh/20MWac

³⁰ <https://globalsolaratlas.info/map?c=11.119117,105.837922,10&s=11.088123,106.066388&m=site>

5.2 Project Emissions

As a solar power generation project, according to ACM0002 (Version 21.0), the project emissions $PE_y = 0$.

5.3 Leakage

According to ACM0002 (Version 21.0), no leakage emissions are considered.

5.4 Estimated Net GHG Emission Reductions and Removals

Net GHG Emission Reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (3)$$

Where

ER_y = Emission reductions in year y (tCO₂e)

BE_y = Baseline emissions in year y (tCO₂e)

PE_y = Project emissions in year y (tCO₂e)

Since $PE_y = 0$,

$$ER_y = BE_y$$

In first crediting period, the total estimated emission reduction of the project is 159,532 tCO₂e, with average annual emission reductions of 22,790 tCO₂e.

The annual emission reductions during first crediting period are listed in below table.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
27-Jan-2022~31-Dec-2022	21,487	0	0	21,487
01-Jan-2023~31-Dec-2023	23,027	0	0	23,027
01-Jan-2024~31-Dec-2024	22,912	0	0	22,912
01-Jan-2025~31-Dec-2025	22,797	0	0	22,797
01-Jan-2026~31-Dec-2026	22,683	0	0	22,683

01-Jan-2027~31-Dec-2027	22,570	0	0	22,570
01-Jan-2028~31-Dec-2028	22,457	0	0	22,457
01-Jan-2029~26-Jan-2029	1,599	0	0	1,599
Total	159,532	0	0	159,532

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07
Source of data	The IFI Dataset of Default Grid Factors version 3.2 Please refer to the section 3.6 for details.
Value applied:	0.5878
Justification of choice of data or description of measurement methods and procedures applied	The IFI Dataset of Default Grid Factors version 3.2. Please refer to the section 3.6 for details.
Purpose of Data	Calculation of baseline emissions
Comments	Fixed in the first crediting period

6.2 Data and Parameters Monitored

Data / Parameter	EG_{facility,y}
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meters
Description of measurement methods and procedures applied	Monitored by electricity meters and monthly recorded by the project operators.
Frequency of monitoring/recording	Monitoring continuously and recording monthly
Value applied:	The ex-ante value please refer to section 5.1, and the parameter will be monitored in the monitoring period.
Monitoring equipment	Electricity meters
QA/QC procedures applied	The data will be cross checked with sale and purchase receipts. The electricity meters will be periodically checked and maintained. Calibration will be carried out in accordance with the PPA. As per the Chapter 7.5 of Meter Accuracy in the PPA, the total metering system accuracy shall be 0.2% or better. Should total metering system accuracy be measured above 0.2%, the project owner will immediately report this to the EDC grid for their inspection and arrange for meters to be tested for accuracy and re-calibrated. All electronic data and paper documents will be archived for two years following the end of the crediting period.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculated as difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid.
Comments	-

6.3 Monitoring Plan

The monitoring plan presented in this report assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final emission reduction of the project. This monitoring plan will be implemented by the project owner during the project operation. The details of the monitoring plan are specified as follows:

Management Structure

The Project owner organizes a specific monitoring team to be responsible for data collection, supervision and witness the whole process of data measuring and recording. Monitoring manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring, recording of electricity meters is to be carried out by designated monitoring staff. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant sales and purchase receipts, and the calculation of the emission reductions.

Data and parameters to be monitored

The parameter to be monitored is Quantity of net electricity generation supplied by the project plant/unit to the grid, which is calculated as difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid.

As shown in Figure 6-1 below, the electricity generation supplied by the project is sent to the 115/22kV Chrak Mtes Substation by the 22kV transmission line, and then delivered to the EDC grid through 115kV transmission line. (a) is to be continuously measured by electricity meters M1 and M2 installed at the H01 STX1 and H03 STX4 of 22kV lines in the low voltage side of the project. (b) is to be continuously measured by electricity meters M1, M2 and M3 installed at the H01 STX1, H03 STX4 and H05 HF of 22kV lines in the low voltage side of the project. Please refer to the section 7.1 for the detailed meters information.

Meter readings will be at least monthly recorded. The accuracy of all electricity meters is not lower than 0.5s.

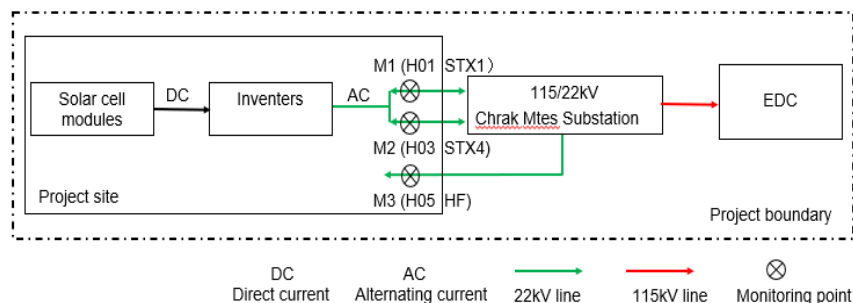


Figure 6-1 The process flow chart of the project

Data collection

Monitoring staffs of the proposed project are responsible for data collection. Designated teams will read and collect the monitored data regularly and recorded in the meter reading records or operation log. The meter readings are also available to the power grid through remote reading system. Monthly settlement for electricity supplied to the grid will be based on the meter readings. Meter reading records will serve as the main data source for emission reductions calculation, after crosschecking with the sales and purchase receipts. All data files, relevant sales receipts will be collected by designated monitoring staff, who will prepare backup in time and archive all documents properly.

Quality assurance

As per Schedule 6 of Metering system, standard and testing in PPA, prior to installation of meters, the project will deliver such meters to the EDC grid's qualified test laboratory for accuracy calibration tests to be carried out to the satisfaction of the EDC grid. After satisfactory accuracy calibration tests, such meters will be sealed and delivered to the project site for installation. After installation of such meters, site tests will be undertaken to confirm the accuracy of the meters. All electricity meters of the project have been approved by the EDC grid and M1, M2 and M3 have been installed at the 22kV lines in the low voltage side of the project, the accuracy of electricity meters is not lower than 0.5s.

The calibration frequency should comply with PPA. As per the Chapter 7.5 of Meter Accuracy in the PPA, the total metering system accuracy shall be 0.2% or better. Should total metering system accuracy be measured above 0.2%, the project owner will immediately report this to the EDC grid for their inspection and arrange for meters to be tested for accuracy and re-calibrated. All electricity meters have been calibrated prior to the project start date. The total metering system accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated.

Applicable International Electrotechnical Commission (IEC) standards such IEC 62052 General requirements, tests and test conditions and IEC 62053 Electricity metering equipment - Particular requirements are applied for the metering system.

Error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

Data file management

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy. Other documents in paper e.g., operation logbooks, will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{\text{facility},y}$												
Data unit	MWh/yr												
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y												
Value applied:	34,694.918												
Comments	This parameter is monitored continuously and recorded monthly. The parameters of $EG_{\text{facility},y}$ is calculated by the difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid. (a) the quantity of electricity supplied by the project plant to the grid is measured continuously by bi-directional electricity meters M1 and M2. (a) is calculated by the sum of these two electricity meters readings. (b) the quantity of electricity delivered to the project plant from the grid is measured continuously by electricity meters M1, M2 and M3. (b) is calculated by the sum of these three electricity meters readings. In this first monitoring period, by cross checking, the monthly meter records of both (a) and (b) were in consistent with the data in the sale and purchase receipts. The accuracy of all electricity meters is not lower than 0.5s. The key information of electricity meters is shown as below. <table><tr><td>Monitoring equipment</td><td>M1</td><td>M2</td><td>M3</td></tr><tr><td>Location</td><td>H01 STX1 (Incoming F1)</td><td>H03 STX4 (Incoming F2)</td><td>H05 HF</td></tr><tr><td>Type</td><td>Mk6E</td><td>Mk6E</td><td>Mk6E</td></tr></table>	Monitoring equipment	M1	M2	M3	Location	H01 STX1 (Incoming F1)	H03 STX4 (Incoming F2)	H05 HF	Type	Mk6E	Mk6E	Mk6E
Monitoring equipment	M1	M2	M3										
Location	H01 STX1 (Incoming F1)	H03 STX4 (Incoming F2)	H05 HF										
Type	Mk6E	Mk6E	Mk6E										

	Serial No	920809277	920809279	250691873
	Accuracy	0.2S	0.2S	0.5S
	Calibration date	14-Jan-2022	14-Jan-2022	24-Jan-2022
The total metering system accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated.				

7.2 Baseline Emissions

According to the methodology ACM0002 (Version 21.0), baseline emissions are determined based on the section 5.1. The BE_y is ex-post determined as the following equations:

$$BE_y = EG_{\text{facility},y} \times EF_{\text{grid},CM,y}$$

$EG_{\text{facility},y}$ is calculated by the difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid. (a) is calculated by the sum of these two electricity meters readings, (b) is calculated by the sum of these three electricity meters readings. The monthly data of (a), (b) and $EG_{\text{facility},y}$ is listed as below.

Table 7-1 Monthly data of (a), (b) and $EG_{\text{facility},y}$

Period	The quantity of electricity supplied by the project plant to the grid (MWh)				The quantity of electricity delivered to the project plant from the grid (MWh)					EG _{facility,y} (MWh)
	Meter reading records		Sale receipts	The quantity of electricity supplied by the project plant to the grid	Meter reading records			Purchase receipts	The quantity of electricity delivered to the project plant from the grid	
	M1 (H01_STX1-Incoming F1)	M2 (H03_STX4-Incoming F2)			M1 (H01_STX1-Incoming F1)	M2 (H03_STX4-Incoming F2)	M3 (H05 HF)			
	A	B			E	F	G			
27-Jan-2022~31-Jan-2022	155.233	159.605	314.838	314.838	1.457	1.427	0.471	3.355	3.355	311.483
01-Feb-2022~28-Feb-2022	1,544.461	1,707.439	3,251.900	3,251.900	7.270	8.221	5.696	21.187	21.187	3,230.713
01-Mar-2022~31-Mar-2022	1,578.349	1,740.052	3,318.401	3,318.401	8.053	9.096	12.383	29.532	29.532	3,288.869
01-Apr-2022~30-Apr-2022	1,485.844	1,645.525	3,131.369	3,131.369	7.740	8.626	4.244	20.610	20.610	3,110.759
01-May-2022~31-May-2022	1,519.180	1,676.232	3,195.412	3,195.412	1.681	1.694	2.777	6.152	6.152	3,189.260
01-Jun-2022~30-Jun-2022	1,580.352	1,758.472	3,338.824	3,338.824	1.164	1.041	2.836	5.041	5.041	3,333.783
01-Jul-2022~31-Jul-2022	1,415.036	1,552.316	2,967.352	2,967.352	0.844	0.771	2.851	4.466	4.466	2,962.886
01-Aug-2022~31-Aug-2022	1,502.149	1,662.012	3,164.161	3,164.161	1.148	1.036	2.881	5.065	5.065	3,159.096
01-Sep-2022~30-Sep-2022	1,395.233	1,533.147	2,928.380	2,928.380	1.502	1.388	2.719	5.609	5.609	2,922.771
01-Oct-2022~31-Oct-2022	1,436.198	1,584.019	3,020.217	3,020.217	1.509	1.362	2.888	5.759	5.759	3,014.458
01-Nov-2022~30-Nov-2022	1,310.661	1,448.030	2,758.691	2,758.691	1.522	1.421	2.731	5.674	5.674	2,753.017
01-Dec-2022~31-Dec-2022	1,626.564	1,797.184	3,423.748	3,423.748	1.512	1.583	2.830	5.925	5.925	3,417.823
Total	16,549.260	18,264.033	34,813.293	34,813.293	35.402	37.666	45.307	118.375	118.375	34,694.918

Based on the section 5.1, the $EF_{grid,CM,y}$ of the project is 0.5878 tCO₂e/MWh, therefore BE_{CH₄} is ex post calculated value during this monitoring period as the follows.

Period	$EG_{facility,y}$	$EF_{grid,CM,y}$	Baseline Emission
	MWh	tCO ₂ e/MWh	tCO ₂ e
	A	B	C=A*B
27-Jan-2022~31-Dec-2022	34,694.918	0.5878	20,393
Total	34,694.918	0.5878	20,393

Therefore, the baseline emission is 20,393 tCO₂e during this first monitoring period.

7.3 Project Emissions

As a solar power generation project, according to ACM0002 (Version 21.0), the project emissions $PE_y = 0$.

7.4 Leakage

According to ACM0002 (Version 21.0), no leakage emissions are considered.

7.5 Net GHG Emission Reductions and Removals

As described in section 7.2, 7.3 and 7.4 above, in this monitoring period from 27-Jan-2022 to 31-Dec-2022,

$$BE_y = 20,393 \text{ tCO}_2\text{e};$$

$$PE_y = 0 \text{ tCO}_2\text{e};$$

$$ER_y = BE_y - PE_y = 20,393 \text{ tCO}_2\text{e}$$

The detail information can be obtained as below:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
Year 2022 (27-Jan-2022~31-Dec-2022)	20,393	0	0	20,393
Total	20,393	0	0	20,393

The project 14,919.983 MWh of total net electricity supplied to the grid and achieved 20,393 tCO₂e emission reductions during this monitoring period.

The achieved emission reductions during this reporting period are only 5.09% lower than the estimation, which is due to normal fluctuation of solar energy and is deemed reasonable. During this monitoring period, there are no special events except for the planned maintenance and overhaul.

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
(27-Jan-2022~31-Dec-2022) 21,487	20,393	-5.09%	The achieved emission reductions during this reporting period are only 5.09% lower than the estimation, which is due to normal fluctuation of solar energy and is deemed reasonable.

APPENDIX 1: <EVIDENCE OF SDG >

Evidence for SDG 7 and SDG 13: From 27-Jan-2022 onwards, quantity of net electricity generation supplied by the project and corresponding emission reduction in the monitoring period have been listed in table below. The emission reduction is a calculation value and please refer to the section 7 for detailed calculation process. $EG_{\text{facility},y}$ is measured by electricity meters and all monthly meter records have been submitted to VVB for review. The part monthly meter records are shown as below.

Period	SDG 7	SDG 13
	$EG_{\text{facility},y}$ (MWh)	ER (tCO ₂)
27-Jan-2022~31-Dec-2022	34,694.918	20,393

A. MONTHLY ENERGY METERS				
Feeder Name	Code	28-Feb 2022	31-Mar 2022	Total
H01 STX1 (Incoming F1)	1.8.0 (Ex kWh)	1699,694	3278043	1,578,349
	3.8.0 (Ex kVarh)	11007	20915	9,908
	2.8.0 (Im kWh)	8727	16789	8,062
	4.8.0 (Im kVarh)	626901	1277087	650,186
H03 STX4 (Incoming F2)	1.8.0 (Ex kWh)	1867044	3607096	1,740,052
	3.8.0 (Ex kVarh)	14207	27270	13,063
	2.8.0 (Im kWh)	9648	18755	9,107
	4.8.0 (Im kVarh)	697103	1411662	714,559
H05 HF	1.8.0 (Imp kWh)	6167	12383	6,216
	3.8.0 (Im kVarh)	0	0	0
	2.8.0 (Ex kWh)	0	0	0
	4.8.0 (Ex kVarh)	2331473	4688145	2,356,672
TOTAL ENERGY EXPORTED TO EDC THROUGH INC F1 & INC F2				3,318,401
TOTAL ENERGY IMPORTED FROM EDC THROUGH INC F1, INC F2 + HF				23,385

Figure 8-1 The monthly meter records in March

A. MONTHLY ENERGY METERS				
Feeder Name	Code	30-Sep 2022	31-Oct 2022	Total
H01 STX1 (Incoming F1)	1.8.0 (Ex kWh)	12175837	13612035	1,436,198
	3.8.0 (Ex kVarh)	35830	36829	999
	2.8.0 (Im kWh)	30868	32377	1,509
	4.8.0 (Im kVarh)	5181537	5818325	636,788
H03 STX4 (Incoming F2)	1.8.0 (Ex kWh)	13434800	15018819	1,584,019
	3.8.0 (Ex kVarh)	46398	47642	1,244
	2.8.0 (Im kWh)	33311	34673	1,362
	4.8.0 (Im kVarh)	5656766	6353257	696,491
H05 HF	1.8.0 (Im kWh)	30691	33579	2,888
	3.8.0 (Im kVarh)	0	0	0
	2.8.0 (Ex kWh)	0	0	0
	4.8.0 (Ex kVarh)	11678234	12765144	1,088,044
TOTAL ENERGY EXPORTED TO EDC THROUGH INC F1 & INC F2				3,020,217
TOTAL ENERGY IMPORTED FROM EDC THROUGH INC F1, INC F2 + HF				5,759

Figure 8-2 The monthly meter records in October

Evidence for SDG 8: The original staff list with more detailed personal information has been submitted as a separate document to VVB due to confidentiality requirement of the project proponent. The simplified staff list is shown as below.

No.	Emp No	Name	Gender	Nationality	Working Information		EDUCATION	
		English Name			Date Join	POSITION	Degree	Magor
1	PCL038	KHENG BUNTHEL	M	CAMBODIA	15-Aug-17	SUBSTATION OPERATOR	Bachelor Degree of Technology	Electrical Engineering
2	PCL048	SEANG CHAMRONG	M	CAMBODIA	11-Sep-17	SUBSTATION OPERATOR	Bachelor Degree of Technology	Electrical Engineering
3	PCL061	YON SOKHACH	M	CAMBODIA	02-Jan-18	SUBSTATION OPERATOR	Bachelor Degree of Technology	Electrical Engineering
4	PCL077	KHUN DARA	M	CAMBODIA	01-Jul-18	SUBSTATION OPERATOR	Bachelor Degree of Technology	Electrical Engineering
5	PCL146	LIM DEVIT	M	CAMBODIA	01-Feb-22	SOLAR PLANT OPERATOR	Bachelor Degree of Technology (Year 4)	Electrical Engineering
6	PCL147	KHENG BUNRITH	M	CAMBODIA	01-Feb-22	SOLAR PLANT OPERATOR	Bachelor Degree of Technology (Year 3)	Electrical Engineering
7	PCL148	TOCH CHANTHORN	F	CAMBODIA	01-Feb-22	GENERAL WORKER	Primary School	-

Figure 8-3 The simplified staff list